

Challenge Overview

- **Category:** Git / Forensics
- **Description:** The challenge hints that a file named `secrets.txt` contains critical information, but inspecting it normally reveals only a dummy placeholder. The commit message gives a major clue: `//This file seems suspicious... maybe there's a way I can see if anything has replaced it...?`

The Twist: Git Replace Mechanism

Git features a plumbing command called `git replace`. It allows developers to tell Git to transparently substitute one object (a commit, tree, or blob) with another without rewriting the repository history or changing commit hashes.

Crucially, **Git does not fetch replacement references (`refs/replace/*`) by default** when you run a standard `git clone` or `git fetch`. This means the flag was sitting on the remote server, mapped as a replacement object, while your local repository initially only saw the dummy decoy commit.

Step-by-Step Solution

Step 1: Clone and Inspect

Cloning the repository and looking at the log reveals a commit by `zinkozapper` adding `secrets.txt`.

```
Bash
git clone git://chals.cyberjousting.com:1363/challenge
cd challenge
cat secrets.txt
# Output: This is where we've already put our secrets!
```

Step 2: Fetch the Hidden Replacements

Because Git ignores replacement refs during a standard clone, you have to explicitly pull them from the remote server:

```
Bash
git fetch origin 'refs/replace/*:refs/replace/*'
```

Step 3: Verify and Read the Flag

Once the references are downloaded, Git immediately maps them. Listing the active replacements confirms that the commit hash `f88c2adbe257...` has an active substitute.

Running `git show` now forces Git to render the newly fetched replacement object instead of the decoy:

Bash

command `git show f88c2adbe25705ad54cabfd0c84826235446ffd0`

Output

Diff

commit `f88c2adbe25705ad54cabfd0c84826235446ffd0` (HEAD -> main, replaced, origin/main, origin/HEAD)

Author: zinkozapper <zinkogamez@gmail.com>

diff --git a/secrets.txt b/secrets.txt

new file mode 100644

index 0000000..05a7982

--- /dev/null

+++ b/secrets.txt

@@ -0,0 +1 @@

+byuctf{I_lov3_s3cr3t_files}

Flag

`byuctf{I_lov3_s3cr3t_files}`

```
diff --git a/secrets.txt b/secrets.txt
new file mode 100644
index 0000000..05a7982
--- /dev/null
+++ b/secrets.txt
@@ -0,0 +1 @@
+byuctf{I_lov3_s3cr3t_files}
byuctf{I_lov3_s3cr3t_files}
tree 5e5ac8713063fd6a8f0f3a9b5e64ff971dc14992
parent 4f4bc00eb57ae05fe351548f9b4e48d3fae7f268
author zinkozapper <zinkogamez@gmail.com> 1773251168 -0600
committer zinkozapper <zinkogamez@gmail.com> 1773251168 -0600

Added secrets.txt

//This file seems suspicious... maybe there's a way I can see if anything has replaced it
..?

(kali㉿kali)-[~/Desktop/ctf/git/challenge]
$
```